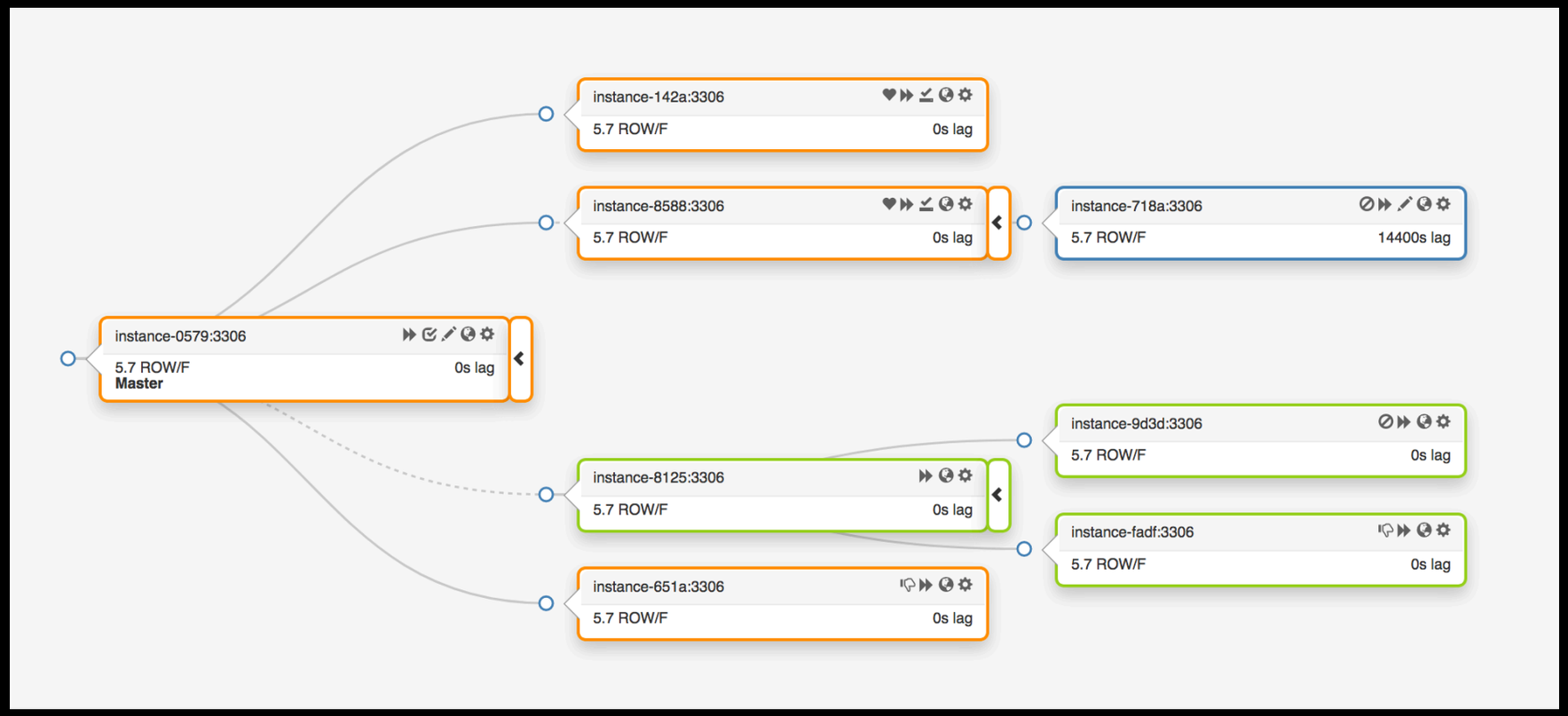
MySQL High Availability at GitHub

老的架构：DNS+VIP+orchestrator

1.客户端通过域名访问mysql，例如：mysql-writer-1.github.net
2.域名对应master的VIP，master切换时需要把VIP从旧的master更新到新master，先让旧master放弃当前VIP，并设置新master的VIP

主要问题：
1. split-brain：旧master死机重新起来仍使用旧的VIP
2. master替换后指向旧master的连接可能仍然在使用
3. VIP切换只能在同一个路由器或者交换机下面，无法跨机房
4. DNS修改的生效需要时间间隔



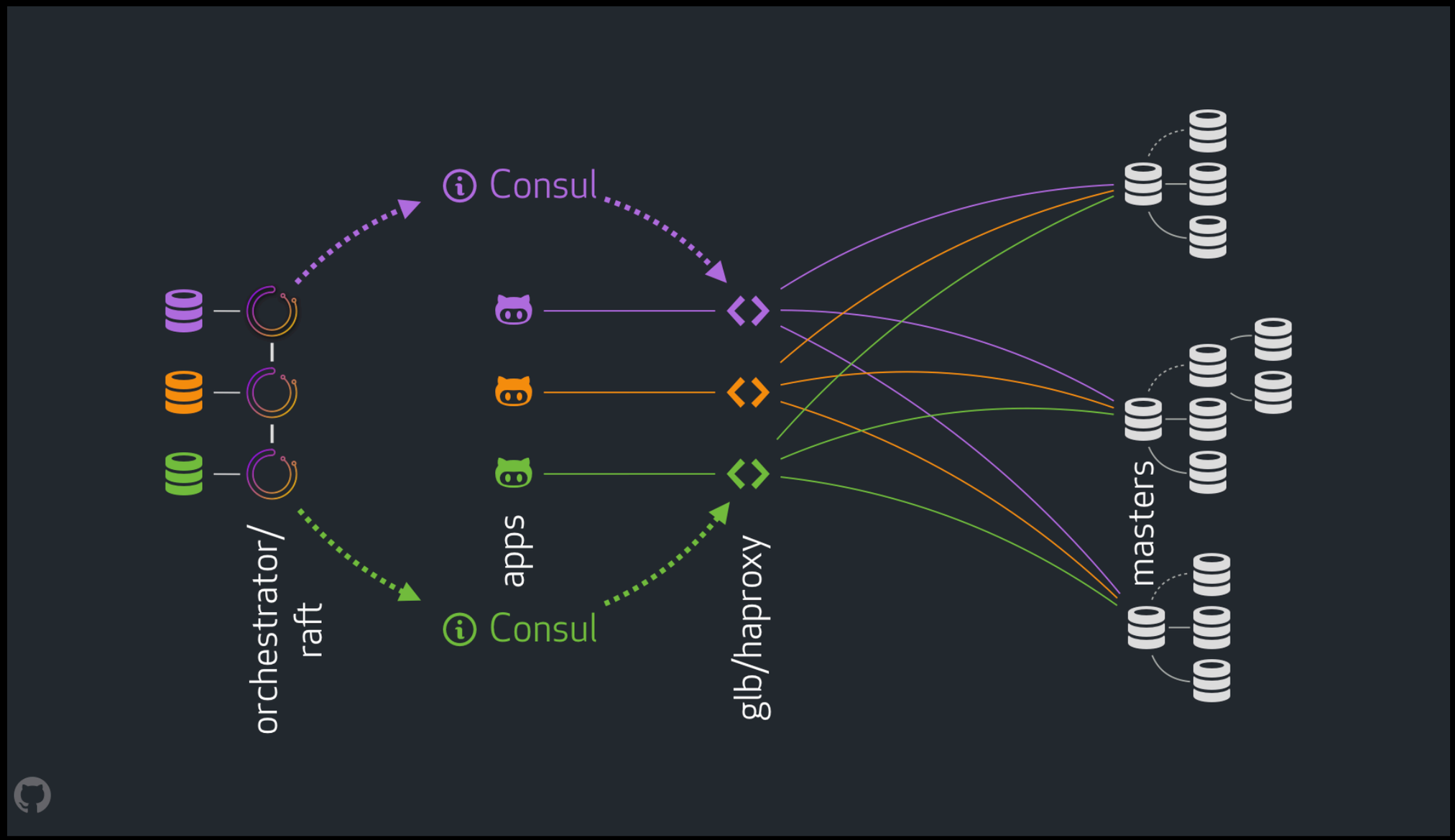
DNS+VIP 架构图

target of new solution

1. master故障自动检测和替换，旧master的连接尽快关闭
2. 减少故障时间
3. 支持DC级别故障
4. 减少故障导致的数据丢失
5. 减少VIP替换等不可控操作

HA solution: orchestrator/raft, Consul, GLB

1. orchestrator/raft: failure detection and advertise
2. Consul: KV store用，存储master信息
3. GLB：接入层，同机部署Consul-Template, 监听consul的更新



新架构

orchestrator/raft

1. Orchestrator负责死机探测，每个机房部署一台或2台
2. 多个Orchestrator使用raft协议达成一致
3. failover由raft的leader操作，并且通过raft协议同步到其他orchestrator
4. orchestrator将master变更更新到consul

master故障处理流程：

1.orchestrator检测到master故障，通知leader
2.raft的leader重新选master，并将master变更信息同步到其他orchestrator
3.orchestrator向consul更新master信息
4.consul-template检测到consul更新后,通知GLB，GLB负责将请求路由到新master

MySQL相关

1. Semi-synchronous replication — 减少故障引起的数据丢失
   master接受请求后将请求复制到一个或两个副本后返回前端
2. mysql状态变更：read-write, writeable
3. mysql的复制使用Pseudo GTID
   Pseudo GTID is the method of injecting unique entries into the binary logs, such that they can be used to match/sync replicas without direct connection, or replicas whose master is corrupted/dead.

Drawback

1.consul和orcheastrator同时故障，影响系统可用性

2.GLB负责接入用户请求，master无法感知client信息

3.split-brain
master在isolated DC时，其他DC会重新选取master，影响数据一致性

github10月21的故障就是由于split-brain问题引起。https://blog.github.com/2018-10-30-oct21-post-incident-analysis/

最近看的好文章

① 负载均衡
- https://medium.com/netflix-techblog/netflix-edge-load-balancing-695308b5548c
- https://code.fb.com/data-infrastructure/dhcplb-an-open-source-load-balancer/
- :https://githubengineering.com/glb-director-open-source-load-balancer

② facebook配置变更：https://code.fb.com/data-infrastructure/location-aware-distribution-configuring-servers-at-scale/

③ netflix的cache演进：https://medium.com/netflix-techblog/evolution-of-application-data-caching-from-ram-to-ssd-a33d6fa7a690

④ mysql高可用性方案：https://githubengineering.com/mysql-high-availability-at-github/

⑤ goroutine调度简介：https://tonybai.com/2017/06/23/an-intro-about-goroutine-scheduler/

⑥ 存储方面
- https://code.fb.com/core-data/under-the-hood-facebook-s-cold-storage-system/
- https://code.fb.com/core-data/akkio/

⑦ Uber的多DC存储系统：
- https://eng.uber.com/herb-datacenter-replication/
- https://eng.uber.com/schemaless-part-one/

谢谢！